

Rational logarithmic extensions of the minimal models and their simple modules

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Outline

- 1 Review of VOAs
- 2 The triplet algebra and its simple modules
- 3 Indecomposable modules and tensor products

Review of VOA definitions

Vertex operator algebras (VOAs) are axiomatisations of the chiral algebra of conformal field theories [Borcherds], [Frenkel, Lepowsky, Meurman].

Def: A vertex operator algebra is a graded vector space

$$V = \bigoplus_{n \geq 0} V[n] .$$

- $\dim V[n] < \infty$, $\dim V[0] = 1$.
- Vacuum vector $|0\rangle \in V[0]$ and conformal vector $T \in V[2]$.
- For $A \in V[h_A]$,

$$Y(A; z) = \sum_{n \in \mathbb{Z}} A_n z^{-n-h_A} \in \text{End}(V)[[z, z^{-1}]] ,$$

$$Y(T; z) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2} ,$$

$$[L_m, L_n] = (m - n)L_{m+n} + \delta_{m+n,0} m(m^2 - 1) \frac{c_V}{12} .$$

Review of VOA definitions

VOAs satisfy the axioms:

- $Y(|0\rangle; z) = \text{id}.$
- $V[n] = \{A \in V | L_0 A = nA\}.$
- Operator state correspondence: $Y(A; z)|0\rangle = A + zV[[z]].$
- For $A \in V$, $\partial_z Y(A; z) = Y(L_{-1}A; z).$
- Locality: For $A, B \in V$ there exists an $N > 0$ such that $(z - w)^N [Y(A; z), Y(B; w)] = 0.$
- Operator product expansion: For $A \in V[h_A], B \in V$

$$\begin{aligned} Y(A; z)Y(B; w) &= Y(Y(A; z - w)B; w) \\ &= \sum_{n \in \mathbb{Z}} Y(A_n B; w)(z - w)^{-n - h_A}. \end{aligned}$$

Review of VOA definitions

Def: A V -module M is a $\mathbb{C}$ graded vector space

$$M = \bigoplus_{h \in \mathbb{C}} M[h]$$

which carries a representation $Y_M(\ ; z)$ such that:

- The $M[h]$ are generalised L_0 eigenspaces with generalised eigenvalue h .
- For fixed $\text{im}(h)$, $\dim M[h] = 0$ for $\text{re}(h) \ll 0$
- For $A, B \in V$ and $C \in M$

$$Y_M(|0\rangle; z) = \text{id}_M$$

$$Y_M(A; z)C \in M((z))$$

$$Y_M(A; z)Y_M(B; w)C = Y_M(Y(A; z - w)B; w)C$$

Overview of $\mathcal{M}_{p_+, p_-}$

This talk is about the extended W -algebra of type $\mathfrak{sl}_2$ at positive rational level, also called the triplet algebra. We denote this VOA by $\mathcal{M}_{p_+, p_-}$.

- The VOA $\mathcal{M}_{p_+, p_-}$ is given by the kernels of 2 screening operators S_+, S_- on a lattice VOA $\mathcal{V}_{p_+, p_-}$ [Feigin, Gainutdinov, Semikhatov, Tipunin].
- The central charge of $\mathcal{M}_{p_+, p_-}$ is the central charge of the minimal models

$$c_{p_+, p_-} = 1 - 6 \frac{(p_+ - p_-)^2}{p_+ p_-}$$

- As a module over itself $\mathcal{M}_{p_+, p_-}$ is reducible

$$0 \rightarrow \mathcal{X}_{1,1}^+ \longrightarrow \mathcal{M}_{p_+, p_-} \longrightarrow L(0, c_{p_+, p_-}) \rightarrow 0$$

- As a Virasoro module $\mathcal{M}_{p_+, p_-}$ decomposes into an ∞ direct sum of Virasoro modules. All but one of these modules are simple.

Overview of $\mathcal{M}_{p_+, p_-}$

- The $\mathcal{M}_{p_+, p_-}$ admit logarithmic modules, that is, there exist $\mathcal{M}_{p_+, p_-}$ -modules on which the action of L_0 is non-diagonalisable.
- The algebra of screening operators forms a quantum group $\mathfrak{g}_{p_+, p_-}$.
- The two categories $\text{Rep}(\mathcal{M}_{p_+, p_-})$ and $\text{Rep}(\mathfrak{g}_{p_+, p_-})$ are related but not equivalent as either tensor or abelian categories.
- Zhu's algebra for $\mathcal{M}_{p_+, p_-}$ is finite-dimensional. Both algebras admit $\frac{(p_+ - 1)(p_- - 1)}{2} + 2p_+ p_-$ simple modules [Adamović, Milas], [Tsuchiya, SW].

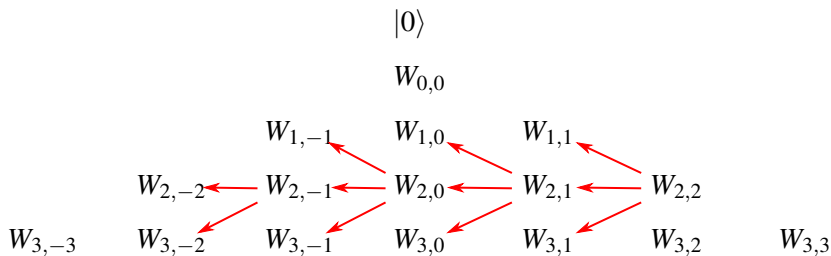
Overview of $\mathcal{M}_{p_+, p_-}$

Thm: The decomposition of $\mathcal{M}_{p_+, p_-}$ in terms of Virasoro highest weight vectors is

$$\begin{array}{ccccccc} & & & & & & |0\rangle \\ & & & & & & W_{0,0} \\ & & & & & W_{1,-1} & W_{1,0} & W_{1,1} \\ & & & W_{2,-2} & W_{2,-1} & W_{2,0} & W_{2,1} & W_{2,2} \\ & \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{array}$$

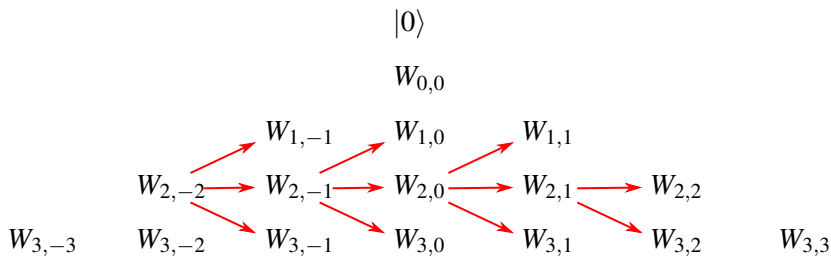
Overview of $\mathcal{M}_{p_+, p_-}$

Thm: The VOA $\mathcal{M}_{p_+, p_-}$ is generated by $T(z)$, $Y(W_{1,-1}; z)$, $Y(W_{1,1}; z)$



Overview of $\mathcal{M}_{p_+, p_-}$

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Overview of $\mathcal{M}_{p_+, p_-}$

Thm: The VOA $\mathcal{M}_{p_+, p_-}$ admits derivations called Frobenius homomorphisms $E, F \in \text{End}(\mathcal{M}_{p_+, p_-})$, that satisfy:

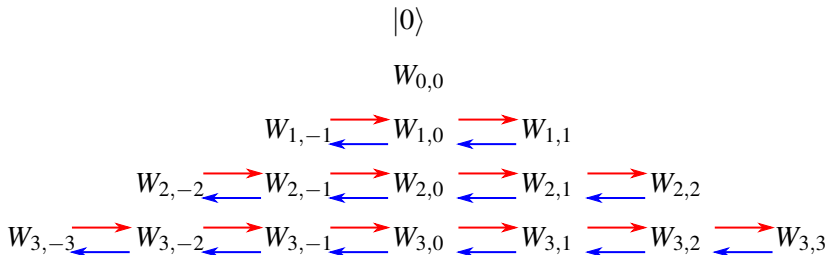
- $[E, L_n] = 0 = [F, L_n], n \in \mathbb{Z}$
- For $A \in \mathcal{M}_{p_+, p_-}$

$$[E, Y(A; z)] = Y(EA; z)$$

$$[F, Y(A; z)] = Y(FA; z)$$

Overview of $\mathcal{M}_{p_+, p_-}$

The action of the Frobenius homomorphisms E, F on $\mathcal{M}_{p_+, p_-}$ is given by:



Overview of $\mathcal{M}_{p_+, p_-}$

Thm: The VOA $\mathcal{M}_{p_+, p_-}$ admits $\frac{(p_+-1)(p_--1)}{2} + 2p_+p_-$ simple modules.

- Minimal model modules $L_{r,s} \cong L_{p_+-r, p_--s}$, $1 \leq r < p_+$, $1 \leq s < p_-$
- Modules with 1-dim space of ground states $\mathcal{X}_{r,s}^+$,
 $1 \leq r \leq p_+$, $1 \leq s \leq p_-$
- Modules with 2-dim space of ground states $\mathcal{X}_{r,s}^-$,
 $1 \leq r \leq p_+$, $1 \leq s \leq p_-$



Tools of the trade

How does one prove these properties?

- Screening operators
- Formal deformations of the conformal structure
- Using a connection between free field theories and symmetric polynomials to apply the theory of Jack symmetric polynomials

The construction of $\mathcal{M}_{p_+, p_-}$

Summary of the construction:

- Define the free field (a.k.a. Heisenberg) VOA $\mathcal{F}_{\alpha_0}$
- Define an extension $\mathcal{V}_{p_+, p_-}$ of $\mathcal{F}_{\alpha_0}$ called a lattice VOA.
- Identify screening operators.
- Define $\mathcal{M}_{p_+, p_-}$ as a subVOA of $\mathcal{V}_{p_+, p_-}$ in terms of screening operators.

The construction of $\mathcal{M}_{p_+, p_-}$

Def Heisenberg VOA $\mathcal{F}_{\alpha_0}, \alpha_0 \in \mathbb{C}$

- Generated by free boson field $b(z) = \sum_{n \in \mathbb{Z}} b_n z^{-n-1}$

$$[b_m, b_n] = m\delta_{m, -n} \cdot 1$$

- $b(z)b(w) = \frac{1}{(z-w)^2} + \dots$
- Energy momentum tensor

$$T(z) = \frac{1}{2} : b(z)^2 : + \frac{\alpha_0}{2} \partial b(z)$$
$$c = 1 - 3\alpha_0^2$$

The construction of $\mathcal{M}_{p_+, p_-}$

Simple $\mathcal{F}_{\alpha_0}$ -modules are given by Fock modules $F_\lambda, \lambda \in \mathbb{C}$.

Def The Fock module $F_\lambda, \lambda \in \mathbb{C}$ is generated by a state $|\lambda\rangle$ such that

- $b_n|\lambda\rangle = \delta_{n,0}\lambda|\lambda\rangle, n \geq 0$
- $F_\lambda \cong \mathbb{C}[b_{-1}, b_{-2}, \dots]|\lambda\rangle$

Rmk $\mathcal{F}_{\alpha_0} \cong F_0$

Prop The category of $\mathcal{F}_{\alpha_0}$ -modules is semisimple.

- Simple objects are given by $F_\lambda, \lambda \in \mathbb{C}$
- The conformal weight of $|\lambda\rangle \in F_\lambda$ is

$$h_\lambda = \frac{1}{2}\lambda(\lambda - \alpha_0)$$
$$L_0|\lambda\rangle = h_\lambda|\lambda\rangle$$

Extending to the lattice

The lattice VOA $\mathcal{V}_{p_+, p_-}$ is an extension of $\mathcal{F}_{\alpha_0}$ at special values of α_0 .
Let $p_+, p_- \geq 2$, $(p_+, p_-) = 1$

$$\alpha_0 = \frac{2p_- - 2p_+}{\sqrt{2p_+p_-}}$$
$$Y = \mathbb{Z}\sqrt{2p_+p_-}$$
$$X = \mathbb{Z}\frac{1}{\sqrt{2p_+p_-}}$$

and let

$$\phi(z) = \hat{b} + b_0 \log z - \sum_{n \neq 0} \frac{b_n}{n} z^{-n}$$
$$b(z) = \partial \phi(z).$$

Extending to the lattice

Def The lattice VOA $\mathcal{V}_{p_+, p_-}$ is given by

$$\mathcal{V}_{p_+, p_-} = \bigoplus_{\lambda \in Y} F_\lambda = \bigoplus_{n \in \mathbb{Z}} F_{n\sqrt{2p_+ + p_-}} \supset \mathcal{F}_{\alpha_0}$$

with fields $T(z)$ and $b(z)$ as in $\mathcal{F}_{\alpha_0}$, $Y(|\lambda\rangle; z) = V_\lambda(z) =: e^{\lambda\phi(z)}$:

$$b(z)V_\lambda(w) = \frac{\lambda}{z-w}V_\lambda(w) + \dots$$

$$V_\lambda(z)V_\mu(w) = (z-w)^{\lambda\mu} : V_\lambda(z)V_\mu(w) :$$

$$T(z)V_\lambda(w) = \frac{\frac{1}{2}\lambda(\lambda - \alpha_0)}{(z-w)^2}V_\lambda(w) + \frac{1}{z-w}\partial V_\lambda(w) + \dots$$

Recall $Y = \mathbb{Z}\sqrt{2p_+p_-}$, $X = \mathbb{Z}\frac{1}{\sqrt{2p_+p_-}}$.

Prop

- The central charge of $\mathcal{V}_{p_+,p_-}$

$$c_{p_+,p_-} = 1 - 6 \frac{(p_+ - p_-)^2}{p_+p_-}$$

is the same as for the minimal models.

- The category of $\mathcal{V}_{p_+,p_-}$ -modules is semisimple. There are $2p_+p_-$ simple objects.

$$V_{[\mu]} = \bigoplus_{\lambda \in \mu + Y} F_\lambda, \quad \mu \in X.$$

The simple modules of $\mathcal{V}_{p_+,p_-}$ are parametrised by the classes of X/Y .

Let $\alpha_{\pm} = \pm\sqrt{2p_{\mp}/p_{\pm}} \in X$. The conformal weight of $V_{\alpha_{\pm}}(w) = S_{\pm}(w)$ is 1. These fields satisfy

$$\begin{aligned} T(z)S_{\pm}(w) &= \frac{1}{(z-w)^2}S_{\pm}(w) + \frac{1}{z-w}\partial S_{\pm}(w) + \dots \\ &= \partial_w \frac{S_{\pm}(w)}{z-w} + \dots \end{aligned}$$

Prop The $S_{\pm}(z)$ are so-called screening operators, that is,

$$\begin{aligned} S_{\pm} &= \text{Res}_{w=0} S_{\pm}(w)dw : \mathcal{V}_{p_+,p_-} \rightarrow V_{[\alpha_{\pm}]} \\ [L_n, S_{\pm}] &= 0, \quad n \in \mathbb{Z} \end{aligned}$$

Def The VOA $\mathcal{M}_{p_+,p_-}$ is given by

$$\mathcal{M}_{p_+,p_-} = \ker(S_+ : \mathcal{V}_{p_+,p_-} \rightarrow V_{[\alpha_+]}) \cap \ker(S_- : \mathcal{V}_{p_+,p_-} \rightarrow V_{[\alpha_-]})$$

Summary and next steps

Summary:

- The VOA $\mathcal{M}_{p_+, p_-}$ is a subVOA of the lattice VOA $\mathcal{V}_{p_+, p_-}$.
- We have constructed $\mathcal{M}_{p_+, p_-}$ as the intersection of kernels of screening operators.

How to proceed:

- In order to prove properties of $\mathcal{M}_{p_+, p_-}$, one needs the screening operator formalism.
- The screening operator formalism involves integrating products of screening operators over certain cycles. Determining these cycles at the minimal model central charge $c_{p_+, p_-} = 1 - 6(p_+ - p_-)^2 / p_+ p_-$ is a very hard problem.
- By deforming the central charge in a deformation parameter ε , one can work at generic central charge.
- By setting $\varepsilon = 0$ one can specialise results obtained at generic central charge.

Screening operators yield explicit formulae for all the Virasoro highest weight vectors in the decomposition of $\mathcal{M}_{p_+, p_-}$ from before:

$$\begin{array}{ccccccc}
 & & & & |0\rangle & & \\
 & & & & W_{0,0} & & \\
 & & & W_{1,-1} & W_{1,0} & W_{1,1} & \\
 & & W_{2,-2} & W_{2,-1} & W_{2,0} & W_{2,1} & W_{2,2} \\
 \cdot & \cdot & \cdot & & & & \cdot & \cdot & \cdot
 \end{array}$$

Screening operators yield explicit formulae for all the Virasoro highest weight vectors in the decomposition of $\mathcal{M}_{p_+, p_-}$ from before:

$$\begin{array}{ccccccccc}
 & & & & & & & & |0\rangle \\
 & & & & & & & & S_+^{[p_+-1]}|\alpha_0^+\rangle \\
 & & & & & & & & S_+^{[p_+-1]}|\alpha_1^+\rangle & S_+^{[2p_+-1]}|\alpha_1^+\rangle & S_+^{[3p_+-1]}|\alpha_1^+\rangle \\
 & & & & & & & & S_+^{[p_+-1]}|\alpha_2^+\rangle & S_+^{[2p_+-1]}|\alpha_2^+\rangle & S_+^{[3p_+-1]}|\alpha_2^+\rangle & S_+^{[4p_+-1]}|\alpha_2^+\rangle & S_+^{[5p_+-1]}|\alpha_2^+\rangle
 \end{array}$$

Screening operators yield explicit formulae for all the Virasoro highest weight vectors in the decomposition of $\mathcal{M}_{p_+, p_-}$ from before:

$$\begin{array}{ccccccc}
 & & & & & & |0\rangle \\
 & & & & & & S_-^{[p_- - 1]} |\alpha_0^- \rangle \\
 & & & & & S_-^{[3p_- - 1]} |\alpha_1^- \rangle & S_-^{[2p_- - 1]} |\alpha_1^- \rangle & S_-^{[p_- - 1]} |\alpha_1^- \rangle \\
 S_-^{[5p_- - 1]} |\alpha_2^- \rangle & S_-^{[4p_- - 1]} |\alpha_2^- \rangle & S_-^{[3p_- - 1]} |\alpha_2^- \rangle & S_-^{[2p_- - 1]} |\alpha_2^- \rangle & S_-^{[p_- - 1]} |\alpha_2^- \rangle
 \end{array}$$

VOA representation theory

- Zhu's algebra $A_0(V)$ is an associative $\mathbb{C}$ algebra that can be constructed from a given VOA V [Zhu], [Frenkel, Zhu].
- Zhu's algebra encodes the action of zero modes on highest weight states [Zhu], [Nagatomo, Tsuchiya].
- There exists a 1-1 correspondence between finite-dimensional simple $A_0(V)$ -modules and simple V -modules [Zhu].
- In practice singular vectors of the VOA are used to determine the centre of Zhu's algebra and classify its representation theory.

The centre of Zhu's algebra and representation theory

The centre of Zhu's algebra is presented by $\mathbb{C}[h]/\langle g(h) \rangle$

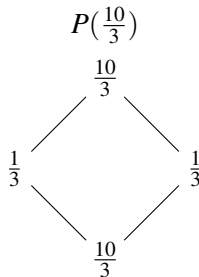
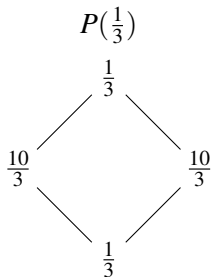
$$\begin{aligned}
 g(h) = & \prod_{[i,j]} (h - \Delta_{i,j})^3 \prod_{i=1}^{p_+-1} \prod_{j=1}^{p_--1} (h - \Delta_{i,j}^+)^2 \prod_{i=1}^{p_+-1} \prod_{j=1}^{p_--1} (h - \Delta_{i,j}^-) \\
 & \prod_{i=1}^{p_+-1} (h - \Delta_{i,p_-}^+)^2 \prod_{i=1}^{p_+-1} (h - \Delta_{i,p_-}^-) \\
 & \prod_{j=1}^{p_--1} (h - \Delta_{p_+,j}^+)^2 \prod_{j=1}^{p_--1} (h - \Delta_{p_+,j}^-) \\
 & (h - \Delta_{p_+,p_-}^+) (h - \Delta_{p_+,p_-}^-)
 \end{aligned}$$

where $\Delta_{r,s}$ is the conformal weight of $L_{r,s}$, and $\Delta_{r,s}^\pm$ are the conformal weights of $\mathcal{X}_{r,s}^\pm$.

The centre of Zhu's algebra and representation theory

$$\prod_{i=1}^{p_+-1} (h - \Delta_{i,p_-}^+) (h - \Delta_{p_+-i,p_-}^-)$$

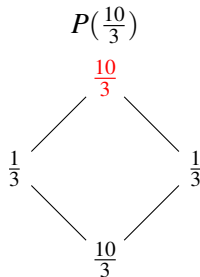
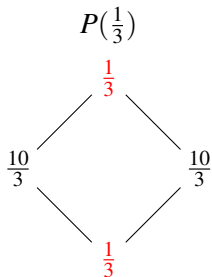
Example: $p_+ = 2, p_- = 3, \Delta_{i,p_-}^+ = \frac{1}{3}, \Delta_{p_+-i,p_-}^- = \frac{10}{3}$



The centre of Zhu's algebra and representation theory

$$\prod_{i=1}^{p_+-1} (h - \Delta_{i,p_-}^+) (h - \Delta_{p_+-i,p_-}^-)$$

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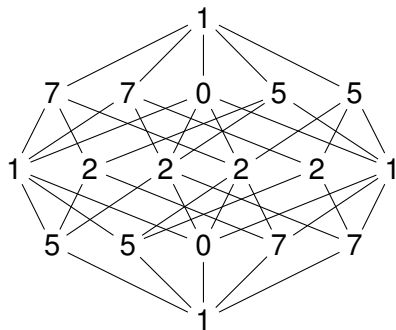
The centre of Zhu's algebra and representation theory

$$\prod_{[i,j]} (h - \Delta_{i,j})^3 (h - \Delta_{i,j}^+)^2 (h - \Delta_{p_+ - i, p_- - j}^+)^2 (h - \Delta_{p_+ - i, j}^-) (h - \Delta_{i, p_- - j}^-)$$

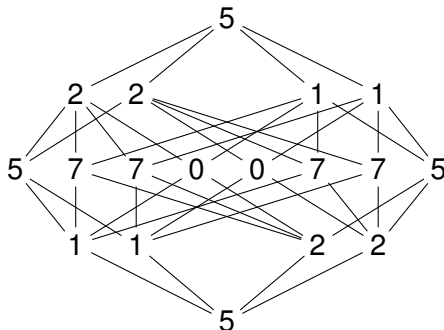
Example: $p_+ = 2, p_- = 3,$

$$\Delta_{i,j} = 0, \Delta_{i,j}^+ = 1, \Delta_{p_+ - i, p_- - j}^+ = 2, \Delta_{p_+ - i, j}^- = 5, \Delta_{i, p_- - j}^- = 7$$

$P(1)$



$P(5)$



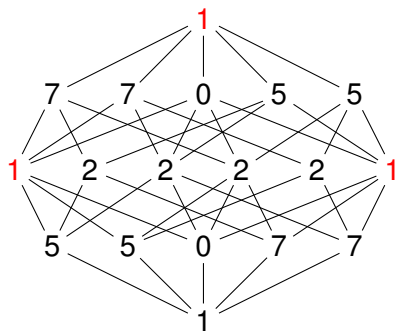
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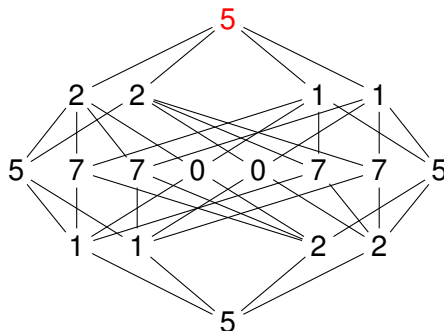
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$P(1)$



$P(5)$



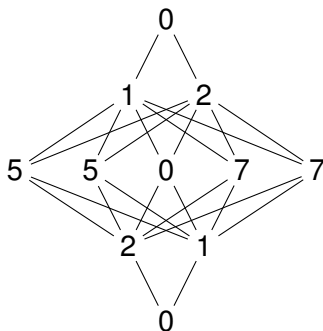
The centre of Zhu's algebra and representation theory

$$\prod_{[i,j]} (h - \Delta_{i,j})^3 (h - \Delta_{i,j}^+)^2 (h - \Delta_{p_+ - i, p_- - j}^+)^2 (h - \Delta_{p_+ - i, j}^-) (h - \Delta_{i, p_- - j}^-)$$

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$P(0)$



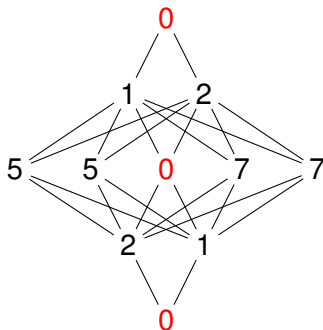
The centre of Zhu's algebra and representation theory

$$\prod_{[i,j]} (h - \Delta_{i,j})^3 (h - \Delta_{i,j}^+)^2 (h - \Delta_{p_+ - i, p_- - j}^+)^2 (h - \Delta_{p_+ - i, j}^-) (h - \Delta_{i, p_- - j}^-)$$

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$$\Delta_{i,j} = 0, \Delta_{i,j}^+ = 1, \Delta_{p_+ - i, p_- - j}^+ = 2, \Delta_{p_+ - i, j}^- = 5, \Delta_{i, p_- - j}^- = 7$$

$P(0)$



Open problems

- Construct projective covers of all simple modules.
- Compute fusion tensor product on $\mathcal{M}_{p_+, p_-}$ -mod.
- Use screening operators to analyse log structure of VOAs on lattices of arbitrary rank finite-dimensional Lie algebras.
- Use screening operators to analyse log structure of affine Kac-Moody superalgebras.

The End

Thank you for your attention!